Interactive Space [ID412]

Professor: Yiyun Kang
TA : Jeanyoon Choi (XD Lab)

Keywords
Space, Interaction | Interactive Media Art | camera Interaction | interactive narrative | real-time graphics | Generative content

Summary of Lecture

Over the past 50 years, interactions between humans, software, machines, and environments have transformed how we create, experience, and understand space. Digital technologies now allow designers and artists to build expressive, participatory, and real-time media environments in which space responds to human action.
This course explores interactive space as the design of relationships between user behaviour, real-time systems, and physical or digital environments. Students will study key critical concepts in interactive media, ranging from embodied interaction and spatial interfaces to camera-based input, data visualisation, generative content, and interactive narrative.
As a final outcome, students will produce a creative project that demonstrates the technical, aesthetic, and conceptual dimensions of interactive space. The goal is to build a meaningful dialogue between user and space through interaction. Students will work both individually and collaboratively throughout the term.

NOTE* Course Prerequisite

1. Taking the 'Immersive Space' course in the spring semester is recommended before taking this class.
2. Priority enrollment is given to undergraduate students majoring in Industrial Design.
3. Graduate students in Industrial Design are also given priority.

Prerequisites: *Participants should understand the basics of HTML, CSS, JavaScript, and React. If you are not familiar with these frontend coding frameworks, it is highly recommended to complete some basic JavaScript tutorials online beforehand.*

During this course, you will learn:

1. Applying methods and technologies for creating audio-visual, interaction-based media environments.
2. Conducting critical and contextual research on interactive projects, with attention to how interactivity connects content, user behaviour, real-time systems, social critique, and physical or digital space.
3. Designing and prototyping creative and speculative real-time interactions that connect user behaviour, computational systems, and physical or digital space.
4. Developing experimental game-engine-based, web-based, and vision-based interactive content for monitor, projection, and spatial installation settings.

Software for teaching

Unreal Engine, Web Technologies (Javascript, React.js, Next.js), Vibecoding Applications (Codex, Claude Code), etc

Course Assignments

ASSSIGNMENT	**OBSERVATION**	Individual	• **Topic:** What is your favourite interactive space project? • **Format:** Reference research and presentation • **Deliverables:** A 15-minute presentation supported by well-organised research, visual materials, and critical analysis.
MID TERM	**GROUP PROJECT**	Group	• **Topic:** Real-time interactive Storytelling • **Format:** Unreal Engine-based interactive prototype • **Tool:** Unreal Engine • **Deliverables:** 15 minutes long live demo with presentation.
FINAL TERM	**SPATIAL AUDIO-VISUAL INSTALLATION**	Individual or Group	• **Topic:** COACTIVITY • **Format:** Interactive audio-visual installation (Sound mandatory). • **Interface:** Custom-built environment • Group or Independent project • **Tool:** Unreal, Web • **Deliverables:** Exhibition-level installation at the 1st-floor Black Room and comprehensive process documentation (archival quality).

Lecture Schedule

WEEK	DATE	ROOM	AGENDA	NOTE	ASSIGNMENT
1	2026 8.31	104	Course Introduction Lecture 1 : Interaction	.	ASSIGNMENT 01 announcement
2	9.07	414	Workshop 1	Unreal 1 - Building Environment	ASSIGNMENT 01 Submission by 12 Sep 11pm
3	9.14	104	ASSIGNMENT 01 presentation		MIDTERM announcement
4	9.21	414	Workshop 2	Unreal 2 - Implementing Interaction	
5	9.28	414	Workshop 3 Tutorial 1	Unreal 3 - Using Data Midterm Concept Tutorial	
6	10.05	.	개천절 대체 공휴일	No Class	
7	10.12	420	Mid-term Tutorial 2	Midterm Development Tutorial	MIDTERM Submission by 18 October 11pm
8	10.19	414	MIDTERM presentation	.	
9	10.26	104	Workshop 4	Web 1: Creative Coding (Canvas, Three.js)	Final project team building Web Assignment 1: Environment Setup & Clone code your favourite website
10	11.02	104	Final Project Presentation 1 Tutorial 3	Final Project Concept Presentation	
11	11.09	104	Workshop 5	Web 2: Cross-Device Interaction (WebSocket)	Web Assignment 2: Build Creative Coding Experience

12	11.16	104	Workshop 6	Web 3: Integrating System (API, Three.js, Mobile, Unreal)	Web Assignment 3: Build Mobile-to-Laptop Interactive Experience
13	11.23	420	Tutorial 4	Final Project Development Tutorial	
14	11.30	104	Final Project Presentation 2	Work in Progress Presentation	
15	12.07	114	Tutorial 5	FInal Setup Presentation	
16	12.14	114	Final Group presentation /	1st floor gallery space	Final material submission (Documentation, Techrider, etc) by 22 April 10pm

Course Assessment

Attendance + Class Participation: 15 %
Midterm project outcome: 20 %
Final project outcome: 40 %
Presentations: 10 %
Final project documentation material: 15 %

Reading References

- **Donna Haraway,** *"A Manifesto for Cyborgs: Science, Technology, and Socialist Feminism in the 1980s,"* Socialist Review 80 (1985): 65–108.
- **Gilles Deleuze and Félix Guattari,** A Thousand Plateaus: Capitalism and Schizophrenia, trans. Brian Massumi (Minneapolis: University of Minnesota Press, 1987).
- **Bruno Latour,** Reassembling the Social: An Introduction to Actor-Network-Theory (Oxford: Oxford University Press, 2005).
- **Timothy Morton,** Hyperobjects: Philosophy and Ecology after the End of the World (Minneapolis: University of Minnesota Press, 2013).
- **Benjamin H. Bratton,** The Stack: On Software and Sovereignty (Cambridge, MA: MIT Press, 2015).
- **Donna Haraway,** *Staying with the Trouble: Making Kin in the Chthulucene* (Durham: Duke University Press, 2016).
- **Nicolas Bourriaud,** The Exform (London: Verso, 2016).
- **James Bridle,** New Dark Age: Technology and the End of the Future (London: Verso, 2018).
- **Jason W. Moore, ed.,** Anthropocene or Capitalocene? Nature, History, and the Crisis of Capitalism (Oakland: PM Press, 2016).
- **N. Katherine Hayles,** Unthought: The Power of the Cognitive Nonconscious (Chicago: University of Chicago Press, 2017).
- **Brian Christian,** The Alignment Problem: Machine Learning and Human Values (New York: W. W. Norton, 2020).
- **Nicolas Bourriaud,** Inclusions: Aesthetics of the Capitalocene (Berlin: Sternberg Press, 2023).
- **Eliezer Yudkowsky and Nate Soares,** If Anyone Builds It, Everyone Dies: Why Superhuman AI Would Kill Us All (New York: Little, Brown and Company, 2025).

- Bloombert Art+Technoolgy
- SIGGRAPH Art Gallery
- Leonardo Journal (MIT press)